

Problem Set 5

POLS W4732

Fall 2026

Due November 13 before the TA section

Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1. Consider an interaction between a political leader and local official, who has the ability to manipulate the vote tally in an upcoming election. The leader can offer a ‘wage’ scheme $w(\cdot)$ to the local official, who can either (i) reject the proposal, (ii) accept and choose high manipulation effort (e_H), or (iii) accept and choose low manipulation effort (e_L). Wages must be nonnegative. When effort is observable, a contract may condition on both effort and the election outcome. When effort is hidden, it may condition only on the election outcome.

Effort affects the realization of a vote tally, which for simplicity we model as a binary outcome $\pi \in \{0, 1\}$ (where $\pi = 1$ implies that the leader stays in power) as follows: $\Pr\{\pi = 1|e_H\} = .6$, $\Pr\{\pi = 1|e_L\} = .5$. The leader’s payoff is given by $\pi - w$, the official’s payoff is given by $\sqrt{w} - c(e)$, with $c(e_H) = .1$, $c(e_L) = 0$. The official’s reservation payoff is $\bar{u} = .2$. Suppose that effort is observable

1. Solve for the optimal wage scheme when the leader wants to implement effort e_L
2. Solve for the optimal wage scheme when the leader wants to implement effort e_H
3. Solve for the leader’s optimal contract (an effort level and a wage scheme)

Now suppose that effort is not observable.

1. Solve for the optimal wage scheme when the leader wants to implement effort e_L
2. Write down the official’s incentive compatibility condition when the leader wants to implement effort e_H
3. Solve for the leader’s optimal contract (an effort level and a wage scheme)

Exercise 2. An agent is being hired to do a task. If he accepts a contract, he can choose between exerting high effort ($e = \bar{e}$) or exerting low effort ($e = \underline{e}$). The only thing the principal observes is whether the agent succeeded or failed at the task. She offers the agent a contract where the wage offered (w) can be contingent on that outcome, but cannot be contingent on the agent's effort choice. If $e = \bar{e}$, then the probability of the agent succeeding is $\bar{p}$. If $e = \underline{e}$, the probability of the agent succeeding is $\underline{p} < \bar{p}$. The payoff to the principal if the agent succeeds is $\bar{\pi}$. The payoff to the principal if the agent fails is $\underline{\pi} < \bar{\pi}$. The principal is risk neutral, so her total payoff is $\pi - w$. The agent is risk averse. Her payoff is $v(w) - c(e)$, with $v' > 0$, $v'' < 0$ and $c(\bar{e}) > c(\underline{e})$. If the agent turns down the contract, he makes a payoff of $\underline{u}$.

1. Suppose the principal wants the agent to accept the contract and exert high effort. Write down the agent's incentive compatibility and participation constraints.
2. Argue that the principal's optimal wage schedule ($w^*(\underline{\pi}); w^*(\bar{\pi})$), given that she wants to induce high effort, involves both constraints binding—i.e., hold with equality.
3. Suppose the principal wants the agent to accept the contract and exert low effort. Write down a contract that achieves this goal.
4. When does the principal prefer the contract that induces high effort to the contract that induces low effort? (Nota bene: you do not need to fully solve for optimal wage. The same applies to the rest of the exercise.)
5. Show that the expected wage is higher under the contract that induces high effort than under the contract that induces low effort.
6. Show that if $\bar{\pi} - w^*(\bar{\pi}) < \underline{\pi} - w^*(\underline{\pi})$, then the principal prefers to induce low effort. Explain.